ADVANCED STUDENT WORKBOOK

# ADVANCED WATER SAFETY

Health & Physical Education | Year 9-10 | Victorian Curriculum F-10 Version 2.0

7-section advanced unit: data analysis, physiology,
social determinants, prevention strategies,
international comparison, diving & spinal injuries,
and summative assessment task.
Research and evidence-based argument required throughout.

STUDENT NAME ______________________

CLASS ______________

TEACHER ______________

**Advanced Water Safety — Year 9-10**

This unit moves beyond description to analysis, evaluation, and synthesis. You will use real Victorian and global data, argue evidence-based positions, and produce a sophisticated communication product as your assessment task.

## 1 Advanced Data Analysis — Victorian & Global Drowning Trends

**Advanced Context — Key Figures**

- 54 fatal drownings in Victoria 2023-24 — highest in over a decade
- Victoria population approx. 6.9 million — rate = 54 / 69 = 0.78 per 100,000
- Australian national rate: approximately 1.2 per 100,000 (RLSSA)
- Non-fatal drownings: 60% above 10-year average
- CALD community representation: ~40% of deaths — highest proportion ever recorded
- Males: 46 of 54 deaths (85%)
- All 54 deaths at unpatrolled locations — 0 deaths at patrolled locations
- WHO global rate: ~320,000 drowning deaths per year; 90% in low/middle-income countries

Sources: lsv.com.au/research/victorian-drowning-reports/ | royallifesaving.com.au | who.int/news-room/fact-sheets/detail/drowning

### Activity 1.1 — Calculations & Interpretation

Individual · ~15 min

**Q1.1a — Calculation** [2 marks]

Calculate the percentage of Victorian drowning deaths that were male. Show your working. Then compare to the global figure (~80%) and explain what the difference suggests about Victorian male risk-taking near water.

*Working: 46 / 54 x 100 = _______% (vs global 80%)*

**Q1.1b — Rate Comparison** [3 marks]

Victoria's rate = 0.78 per 100,000. National rate = 1.2 per 100,000. Calculate: how many deaths would Victoria expect at the national rate? What does this tell you about Victoria's performance? Show all working.

*Expected deaths = 1.2 x _______ (units of 100,000 in 6.9M) = _______*

**Q1.1c — CALD Data Analysis** [3 marks]

CALD communities = 40% of Victorian drowning deaths but ~28% of population. Are CALD Victorians over-represented, under-represented, or proportionally represented? Show reasoning. Give TWO specific structural reasons (not just "language barriers") for this pattern.

*Over-represented / Under-represented / Proportional (circle). Evidence: ______*

**Q1.1d — Data Visualisation** [3 marks]

Create a bar graph comparing Victorian drowning deaths by location type. Research the figures from the LSV Drowning Report (lsv.com.au/research/victorian-drowning-reports/). Label axes, include a descriptive title, and write a 2-sentence analysis beneath.

*[Attach bar graph here or on a separate page]*

*Analysis:* ______________________________

## 2 Physiology of Drowning & CPR Mechanics

**Deeper Knowledge**

- CPR physiology: Chest compressions generate ~25-30% of normal cardiac output
- 4-minute rule: Brain cells begin dying after 4-6 minutes without oxygenated blood
- Bystander effect: CPR rates higher when individuals feel personally responsible
- Compression rate: 100-120 bpm; depth 5-6cm; allow full chest recoil
- Secondary drowning: pulmonary oedema developing hours after near-drowning — rare but serious
- 1-10-1 cold water rule: ~1 min breathing control, ~10 min useful swimming, up to 1 hr before hypothermia

Sources: stjohnvic.com.au | heartfoundation.org.au

### Activity 2.1 — Physiological Sequence & CPR

Individual · ~15 min

**Q2.1a — Physiological Sequence** [3 marks]
Describe the complete physiological sequence during drowning using all five required terms: submersion, hypoxia, laryngospasm, cardiac arrest, cerebral hypoxia. Explain why this is dangerous even in shallow water.

**Q2.1b — Why Early CPR Matters** [3 marks]
Explain why starting CPR within 4 minutes produces significantly better outcomes than starting at 8 minutes, even if professional help arrives at both times. Use the terms: cardiac output, oxygenated blood, brain cells, irreversible damage.

**Q2.1c — Secondary Drowning Research** [2 marks]
Research secondary drowning (pulmonary oedema post-submersion) at stjohnvic.com.au. Explain what it is, what symptoms to look for, and what should happen after ANY submersion incident — even if the person appears fine.

*Source URL:* ________________________________

# 3 Social Determinants of Drowning Risk

## Activity 3.1 — Social Determinants Analysis

Pairs · ~18 min

**Q3.1a — Social Determinants Table** [4 marks]
For each social determinant below, explain the specific mechanism by which it increases drowning risk, give a Victorian data example, and propose one targeted prevention strategy.

| Social Determinant | Mechanism (how it increases risk) | Victorian data link | Prevention strategy |
|---|---|---|---|
| Gender norms (masculinity) | | | |
| CALD background | | | |
| Socioeconomic status | | | |
| Rural/regional location | | | |

**Q3.1b — School Actions** [3 marks]
Propose THREE specific, practical actions a Victorian school could take to reduce CALD drowning disproportionality. Be specific: who delivers it, where, when, and how?

*Action 1:* ______________________________________________

# 4 Prevention Strategies — Evaluation & Evidence

## Activity 4.1 — Prevention Strategy Evaluation

Individual + Research · ~15 min

**Q4.1a — Strategy Evaluation** [4 marks]
For each strategy, evaluate the evidence strength (Strong/Moderate/Limited), identify the primary target population, and explain the key limitation.

| Strategy | Evidence | Target population | Key limitation |
|---|---|---|---|
| Four-sided pool isolation fencing | | | |
| Expand patrolled beach hours | | | |
| Mandatory swim lessons (children) | | | |
| Warning signs (unpatrolled beaches) | | | |
| CALD community swim programs | | | |

**Q4.1b — Resource Allocation** [4 marks]
If you had $5 million to spend on Victorian drowning prevention, which TWO strategies would you prioritise? Justify each with evidence and acknowledge the trade-offs.

*Strategy 1:* __________ *Justification:* ___________________________________

## 5 International Comparison

### Activity 5.1 — Research & Compare

Individual · ~20 min · Devices required

**Research sites:**

royallifesaving.com.au | rlss.org.uk/education | redcross.org/take-a-class/water-safety | who.int/drowning

| Element | RLSSA (Australia) | RLSS UK | Red Cross (USA) |
|---|---|---|---|
| Main program name | | | |
| Core safety message | | | |
| Secondary school focus? | | | |
| One unique feature | | | |

**Q5.1a — Should Victoria Adopt It?** [3 marks]

Choose ONE element from an international approach NOT currently prominent in Victoria. Argue whether Victoria should adopt it, with evidence cited from at least one primary source with a URL.

*Element: __________ From: __________ Source URL: ________*

## 6 Diving, Spinal Injuries & High-Risk Teenage Behaviour

| | | |
|---|---|---|
| **10%** of all spinal cord injuries in Australia are caused by unsafe diving (AIHW) | **20%** of quadriplegia cases in Australia result from diving injuries | **95%** of aquatic spinal injury patients are male (major Australian review) |
| **75%+** of aquatic spinal injury patients are under 30 years of age | **66%** suffer permanent disability — often quadriplegia | **+34%** increase in drowning among 15-20 year olds post-COVID vs pre-COVID (RLSSA) |

Sources: Royal Life Saving Australia (royallifesaving.com.au) | AIHW Diving Injury Report (aihw.gov.au) | Royal Perth Hospital spinal injury study

### Highest-Risk Age Groups & Locations

Royal Life Saving Australia identifies peer pressure, impulsive behaviour, alcohol, "showing off", jumping from structures, and overconfidence in swimming ability as the major contributors to teenage aquatic injuries and fatalities.

**Highest-risk age groups:**

- Children aged 10-14 — highest incidence of diving-related spinal injuries in Australia
- 15-20 year olds — 34% increase in drowning post-COVID compared with pre-COVID period
- Males under 30 — more than 75% of all aquatic spinal injury patients

**High-risk locations:**

- River bridges, wharves, and jetties; dam walls and irrigation channels
- Rope swings over rivers; rock platforms and waterfalls; tidal beach sandbanks
- Quarry swimming holes; floodways and drainage channels

### Australian Case Studies — Recent Incidents

For discussion & analysis below

| Case Study 1 — Terrigal Wharf (NSW, 2025) | Case Study 2 — Torquay Front Beach (Victoria, 2025) | Case Study 3 — Bawley Beach Jetty (NSW, 2024) |
|---|---|---|
| A teenage boy sustained serious spinal injuries after diving from a wharf at Terrigal into water he believed was deep. Helicopter evacuation to hospital. He was unaware of the reduced water depth beneath the wharf structure. | A teenage boy suffered a serious suspected spinal injury after a diving incident at Torquay Front Beach and was airlifted to hospital. Surf beach sandbanks shift with every tide, making depth at any given point highly unpredictable even at a familiar location. | A young man suffered serious injuries diving from a jetty near Bawley Beach. A sandbank had recently formed beneath the jumping area, dramatically reducing water depth. Prior experience at the location created false confidence. |
| **Key themes:** Unknown depth beneath structures; "It looked deep enough" misconception; hidden shallow zones under wharves; catastrophic injury from a single dive. | **Key themes:** Sandbanks change with every tide; holiday risk-taking; lifelong consequences of one dive; surf beaches are not safe for diving. | **Key themes:** Depth changes even at familiar sites; "I've done it before" false confidence; tidal sediment movement; local knowledge does not equal current safety. |

### Why Inland Waterways Are Australia's Highest-Risk Environment

*Royal Life Saving Australia states: "More people drown in inland waterways than any other location in Australia." Five specific factors explain this:*

**1. Depth changes constantly** — Drought, flooding, irrigation releases, sediment build-up, sandbank movement, and dam level changes all alter depth unpredictably — even between morning and afternoon.

**2. Visibility is poor** — Rocks, submerged trees, debris, logs, shopping trolleys, and branches are frequently invisible beneath turbid (murky) inland water.

**3. Cold water shock** — Many Australian rivers and dams remain cold even in hot weather, triggering panic, muscle cramping, loss of coordination, and drowning within minutes.

**4. Alcohol and group risk-taking** — Inland water incidents frequently involve alcohol, teenage males, group peer pressure, and jumping from bridges — a combination that significantly multiplies risk.

**5. Remote locations delay rescue** — River and dam locations often mean delayed ambulance access, limited phone reception, no lifeguards, and delayed spinal immobilisation — all of which worsen outcomes.

**Core prevention message:**

**"Feet first, first time, every time." Never dive into unknown water. Always check depth right now, today — prior safety at a location does not guarantee current safety.**

## Activity 6.1 — Case Study Analysis

Individual · ~18 min

**Q6.1a** **[3 marks]**
Identify two common factors present across all three Australian case studies. For each factor, explain the specific mechanism by which it contributed to the injury — not just that it was present.

**Q6.1b** **[4 marks]**
The data shows 95% of aquatic spinal injury patients are male. Using the social determinants framework from Section 3, construct a structural explanation for this statistic. Your answer must go beyond "males take more risks" and identify the specific social factors that drive this behaviour pattern.

**Q6.1c** **[3 marks]**
Two-thirds of Australian aquatic spinal injury patients suffer permanent disability. Why does a diving injury at the cervical (neck) spine carry a particularly high risk of permanent quadriplegia? Use the physiological content from Section 2 in your answer.

**Q6.1d — Extension** [3 marks]

The Bawley Beach case illustrates false confidence from prior experience. Argue why prior familiarity with a location may be MORE dangerous than complete unfamiliarity. What specific prevention message should follow from this argument?

## Activity 6.2 — Myth vs Evidence Analysis

Individual · ~15 min

For each myth, write a clear evidence-based refutation. Use specific Australian statistics or mechanisms where possible. Explain WHY the myth is false — do not just assert it.

| Myth | Evidence-based refutation (write your response) |
| --- | --- |
| *"If other people are jumping there, it must be safe."* | |
| *"Clear water means deep water."* | |
| *"I've jumped here before — I know this spot."* | |
| *"Good swimmers are safe from diving injuries."* | |
| *"Jumping feet-first is safe — you won't hit your head."* | |
| *"The jetty is high, so the water must be deep."* | |

## Activity 6.3 — Prevention Strategy Design

Individual · ~15 min

**Target group** [3 marks]
Identify your target group (e.g. "teenage males aged 15-18 at Victorian river swimming holes") and explain in one paragraph why this group faces disproportionate risk — citing at least two statistics from this section.

**Prevention strategy** [3 marks]
Describe your strategy in specific terms: What will you do? Through what channel? Who delivers it? What behaviour change does it target? How will you know if it worked (SMART metric)?

**Limitation** [2 marks]
Identify the most significant limitation of your strategy — and explain what a more complete response would look like if resources were not a constraint.

## 7 Summative Assessment Task

**Assessment Task — Choose ONE option**

All options require: at least 3 sources with URLs, at least 1 Victorian/global statistic, clear target audience/addressee, specific measurable call to action, connection to a social determinant or physiological concept from this unit.

**Option A — Policy Brief (600-800 words)**

Address the Victorian Minister for Sport. Two specific recommendations + evidence + counter-argument + measurable success metric for each.

**Option B — Multimodal Campaign**

Target ONE high-risk group from Victorian data. Campaign name, key message, visual concept, platform, 100-word justification of design choices.

**Option C — School Audit (advanced) ★**

Audit your school's water safety provision. Benchmark against Curriculum 2.0 (VCHPEP126, VCHPEM146). Include priority matrix + 3 recommendations.

### Planning Notes

*Option chosen:* _______ *Target audience / addressee:* ________________

*Central argument / key claim:* ______________________________________

*Statistic I will use (with source URL):* ____________________________

*Source 1 (URL):* ___________________________________________________

*Source 2 (URL):* ___________________________________________________

*Source 3 (URL):* ___________________________________________________